



REASONING THINKING
GRADE: 10TH
PROGRESSION TEST 1ST TERM
TEACHER'S NAME: Eliana Rodríguez – David Amaya

**FIRST
TERM
2025–2026**

Learning objective:

Model 3D structures analyzing their 2D triangular components through trigonometric ratios, sine and cosine rules, and vector theory.

Assessment criteria:

- C1: Expresses degrees into radians and vice-versa.
- C2: Interprets three-figured bearings.
- C3: Uses basic operations and translations with vectors.
- C4: Calculates reflections, rotations, translations, and enlargements of functions.
- C5: Identifies trigonometric ratios in right triangles and in the unit circle.
- C6: Differentiates the conditions for using sine and cosine rules.
- C7: Establishes the conditions for using vectors, Pythagorean theorem or trigonometric ratios, and vectors in the solution of context situations.

Student's name: _____ **Group:** _____ **Date:** _____

C3: Vectors

1. Movement of a minute hand. A 10th grade student is studying how angles are measured in radians using an analog clock as a visual reference. She/He notices that when the minute hand moves from the 12 to the 3, it covers a quarter of the clock face. How many radians does the minute hand move when it goes from 12 to 3?

lance drone takes off from base station E, located at the edge of a large farmland. On its first flight path, it travels 15 km North and 8 km East, represented by the vector

$$\vec{v}_1 = (8) 15.$$

Then, the drone adjusts its course and flies 5 km West and 10 km North, represented by

3. Surveillance drone in farmland. A surveil-

$$\vec{v}_2 = (-) 510.$$